

Aflibercept for Radiation Maculopathy Study

A Prospective, Randomized Clinical Study

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Purpose: To evaluate 2 treatment approaches to intravitreal vascular endothelial growth factor antagonist therapy in radiation maculopathy comparing aflibercept delivered by either a 6-week treatment interval or treat-and-adjust interval.

Design: Randomized, prospective clinical trial.

Methods: Forty consecutive patients were enrolled in an institutional review board-approved clinical trial and randomized to aflibercept treatment via 1 of 2 regimens: (1) fixed, every-6-weeks treatment or (2) variable, treat-and-adjust treatment centered around 6 weeks. All patients had a diagnosis of treated uveal melanoma with documented tumor control. All patients showed visually compromising radiation maculopathy confirmed by a decline in best-corrected visual acuity (BCVA) and spectral-domain (SD) OCT documentation of radiation maculopathy.

Main Outcome Measures: Best-corrected visual acuity and SD OCT central retinal thickness at 1 year.

Results: Thirty-nine of 40 patients completed the trial (97.5%) with 1 year of follow-up. Baseline study entry BCVA was 20/63 and was maintained at 20/62 at study conclusion at 60 weeks (1 year). At baseline, SD OCT mean central retinal thickness was 432 μ m and improved to 294 μ m at 60 weeks ($P < 0.02$). At the study conclusion, 42.5% of eyes (17/40) showed better than 20/50 BCVA, and only 5% of eyes (2/40) showed a BCVA worse than 20/200. In the every-6-weeks interval treatment arm, patients received 9 injections, whereas in the treat-and-adjust study arm, patients received 8.4 injections ($P = 0.88$, not significant). One patient experienced an inflammatory response after aflibercept injection, but this did not occur again for this patient, nor for any other study injections (1/400 injections [0.0025%]). No patients demonstrated endophthalmitis or metastatic disease or died during the study window.

Conclusions: Aflibercept seems to limit vision loss associated with radiation maculopathy. In this randomized, prospective clinical study, no difference was found between a fixed 6-week treatment interval and a variable treat-and-adjust interval because virtually all patients required treatment every 6 weeks and were not able to extend. Remarkably, almost half of all treated patients maintained BCVA of 20/50 or better throughout 1 year of treatment. Aflibercept is effective in treating radiation maculopathy, but requires an ongoing treatment approach. *Ophthalmology Retina* 2019;3:561-566 © 2019 Published by Elsevier Inc. on behalf of the American Academy of Ophthalmology

Radiation treatments, which include external beam radiation, plaque brachytherapy, proton beam radiation, helium ion radiotherapy, and γ knife radiotherapy,¹⁻⁹ may result in long-term damage to the retinal vascular endothelial cells, causing radiation maculopathy in more than 75% of eyes undergoing plaque radiotherapy.^{10,11} Currently, posterior uveal melanomas are treated routinely with either brachytherapy (iodine 125, palladium 103, or ruthenium 106) or charged particle radiotherapy (proton or helium beam). These treatments now achieve local tumor control in more than 95% of eyes and avoid enucleation in more than 98% of eyes. These outstanding treatment results have shifted the focus to preserving visual and anatomic function after targeted radiation therapy. Radiation retinopathy involves apoptosis, cell migration, clotting cascades, increased vascular endothelial permeability, aneurysm formation, telangiectasia, and neovascularization.¹² Other

radiation effects on the eye include cataract formation, radiation optic neuropathy, retinal detachment, neovascular glaucoma, nerve fiber layer infarcts, retinal hemorrhages, and vitreous hemorrhage.^{11,13-15}

Radiation damage can be detected and classified through clinical examination, fluorescein angiography, or OCT.¹⁶⁻¹⁸ High-resolution OCT classification (either swept-source or spectral-domain [SD]) is now the standard in the detection and monitoring of radiation-induced macular edema and can be graded on a 1-to-5 scale.¹⁷ This edema grading scale has also been shown to correlate with foveal thickness and visual acuity (VA).¹⁷ Recent findings using SD OCT to detect intraretinal cystic spaces and photoreceptor loss have demonstrated that these earliest signs of radiation maculopathy manifest as macular edema,¹⁹ with average onset of OCT-evident edema by 12 months, or up to 5